

The Third Block Composes: One-Writer Bridges for Multi-Sensor Conditioning of a Frozen Language Model

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2026-08-09

Abstract

Papers 7–9 closed every merged-writer route to composing sensor bricks on a frozen language model: additive weight-space merges, shared-budget activation merges, and cross-channel superposition all die by behavioral dominance, and paper 10 located the deeper constraint — the mode structure of the joint utterance. This note reports the constructive result those negatives were pointing at. Two structures no prior negative tested — *one writer reading many sensors*, and *presence×presence* joint behavior — turn out to be sufficient: a single hypernetwork bridge that reads two sensors’ features and emits one LoRA delta passes a registered gate at every (pair, seed) cell at 4B (free-run joint-heed 0.75–1.00 against measured floors of 0.000–0.167), extends to *three* sensors at 0.90 joint-heed in every cell of both registered triples — lifting the program’s longest-standing blocker, the third-block rail — and promotes to a 27B host with seven of nine cells passing outright (the creature pair perfect at 1.000) and the two exceptions traced, row by row, to a state collision in the shuffled control rather than to the bridge. Per-lane function is retained everywhere, modality-dropout subsets behave as their subsets, and the teacher-forced fit carries into free decode in every cell — the exact opposite of the self-bridge ladder’s fate on the same host. The practical claim: multi-sensor conditioning of a frozen model is solved at the venues tested by giving the sensors one mouth-piece instead of asking several to share a stage.

1 The negatives that shaped the design

The program has one composition result that always works — selection (paper 8) — and a graveyard of superposition attempts. Paper 7: summed weight-space deltas are provably lossless in the merge algebra and behaviorally useless. Paper 9: two activation packets destroy each other under a shared write budget, and block+packet pairs resolve by dominance, dosed but not fungible. Paper 10: at 4B, even a single model cannot be made to *initiate* an utterance carrying recall and presence at once. Every one of those failures shares two structural features: the writers were trained independently and merged after the fact, and the joint behavior asked for spanned two *modes*. The design here removes both: one bridge is trained *natively* on the joint behavior (no post-hoc merger exists to fight), and both contents are present-tense states of the world (one mode, two facts).

2 The harness

Venues: frozen Qwen3.5-4B (March r4 blocks, the dispatch lanes’ banks and scorers) and frozen Qwen3.6-27B (the August r2 column). Registered pairs: **temporal+vitals** (the creature’s pair),

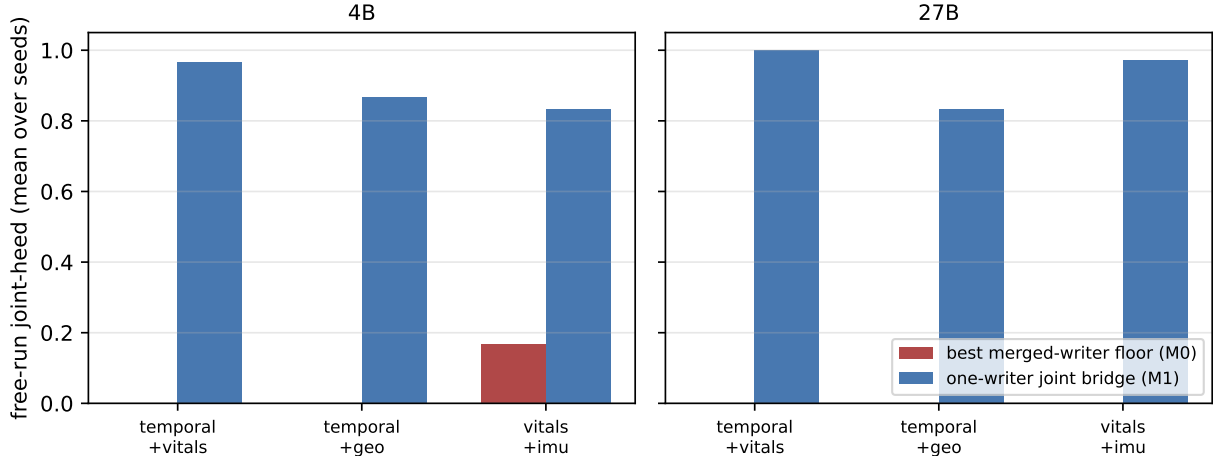


Figure 1: The one-writer result against the merged-writer floors, both venues. Bars: best M0 floor per pair (max over solo, additive merge, prompt-joint, router-oracle) vs the M1 joint bridge’s free-run joint-heed (mean over three seeds, $\beta=1$).

`temporal+geo` (both sinusoidal), `vitals+imu` (mixed encoder families). Joint-heed means both lanes’ registered scorers accept one free-decoded utterance on held-out state pairs, under a neutral situation-naming surface. M0 measured the floors decode-only before anything trained: per-block solo, the paper-7 additive merge re-anchored in-band, a prompt-only joint ask, and an oracle router (paper 8’s ceiling, which surfaces one lane by construction). **Every M0 cell at 27B is 0.000; at 4B everything is 0.000 except one flicker** (the `vitals+imu` merge at 0.167, which does not reproduce at 27B).

The joint bridge is deliberately boring: the house hypernetwork (**BridgeHyper**) with a widened input — two lanes’ 256-d features plus two presence bits — emitting the same flat LoRA delta a single block emits, trained by teacher-forced transcription of machine-validated composed targets (both halves present-tense, four validation rules, each rule tested to bite) with modality dropout 0.25 so one artifact covers both subsets and the joint. Training runs through the same autograd-safe injection path the evaluation uses. Gate per (pair, seed): free-run joint-heed strictly above every M0 cell, per-lane retention ≥ 0.80 of in-band solo, masked inputs reproduce solo behavior, shuffled features at floor, zero-dose bit-exact identity, byte-identical base weights.

3 Results

M1 at 4B: nine of nine cells pass. Table 1 column three carries the headline; the fit column carries the contrast with paper 10 — token accuracy 0.94–0.98 *and the behavior transfers to free decode*, no cell showing the teacher-forced/free-run gap that killed the self-bridge ladder on this same host.

M1b: the third block composes. The same mechanics at $N=3$ (three features, three presence bits, seven dropout subsets): joint-heed — all three lanes’ scorers on one utterance — is **0.90 in every cell of both triples**, the creature’s tuple `temporal+vitals+geo` and the mixed-family `temporal+vitals+imu`, against in-band triple additive-merge floors of 0.000 (Table 2). The third-block rail — the named blocker behind the continuous self-loop, the personality block, and multi-brick serving — lifts at 4B under one writer.

pair	M0 floor	joint by seed	A heed	B heed	fit	passed
temporal+vitals	0.000	0.90/1.00/1.00	1.00–1.00	0.90–1.00	0.94–0.96	3/3
temporal+geo	0.000	0.90/0.80/0.90	1.00–1.00	0.80–0.90	0.97–0.98	3/3
vitals+imu	0.167	0.92/0.75/0.83	0.83–1.00	0.83–1.00	0.95–0.97	3/3

Table 1: M1 at 4B, free-run joint-heed at $\beta=1$ per seed.

triple	pairwise floor	triple merge	joint by seed	passed
temporal+vitals+geo	0.000	0.000	0.90/0.90/0.90	3/3
temporal+vitals+imu	0.167	0.000	0.90/0.90/0.90	3/3

Table 2: M1b at 4B: three sensors, one writer.

Promotion to 27B: seven of nine outright, and a control lesson. `temporal+vitals` is perfect (1.000 at every seed); `vitals+imu` passes everywhere (0.92/1.00/1.00); `temporal+geo` runs 0.80/0.80/0.90 joint with full retention but 2 of its seeds recorded as non-passes on one check: `shuffled_at_floor`. The row-level diagnosis is exact. The shuffle rotated items by one; six of ten held-out pairs share the temporal state under round-robin pairing, and the only joint hits — the same single item at both failing seeds, identical response text — sit on those collisions, against a floor of exactly 0.0. A 27B expresses a partially-right packet where a 4B cannot. The cells stand as recorded (the gate is the gate); the harness’s shuffle is now a state-derangement, and the claim rests on the seven clean cells.

4 What this does and does not establish

It establishes that the merged-writer negatives were about the *merge*, not about multi-sensor conditioning: give N sensors one trained mouth-piece and a frozen host says both — or all three — things at once, message-dependently, with each lane’s function intact, at 4B and at 27B. It also quietly confirms paper 10’s mode account from the constructive side: `presence`×`presence` composes on the exact host where `recall`×`presence` would not compose under any manipulation tested. It does *not* yet attribute the win between “one writer” and “natively-trained joint data” — the registered depth-partition control (M3: independently-trained blocks on disjoint layers, merged) runs next and is designed to separate exactly that pair of explanations; the modality-dropout subsets already show the single writer behaving as each solo when asked to. And N beyond three, other lane families, and serving integration remain open.

5 Reproducibility

Artifacts `multi-sensor-m0-{4b,27b}`, `multi-sensor-m1-{4b,27b}` and `multi-sensor-m1b-4b` are bundled beside this note; every number and the figure render from them alone. Harness: `tracks/multi-sensor/multi_sensor.py`; registrations, results, and the two pre-run amendments in the track’s `PLAN.md`, each dated before its run.

References

- [1] F. Penov. Zero Weight-Space Interference Is Not Enough. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [2] F. Penov. The Escape Is Selectable. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [3] F. Penov. One Stage, One Act. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [4] F. Penov. One Mouth, One Mode. Preprint, 2026. <https://github.com/kortexa-ai/legolm.basis>.
- [5] Qwen Team. Qwen3.5 and Qwen3.6 model families. Model releases, 2025–2026. <https://huggingface.co/Qwen>.